

In-Class Exercise 1 Solutions: Single-Variable Optimization and Logarithms

ECON 320 - Introduction to Mathematical Economics

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Fall 2025

1. Maximization of Profit Function

A firm's total revenue and cost functions are:

$$R(Q) = 80Q - 2Q^2, \quad C(Q) = 20 + 10Q$$

Solution:

(a) *Profit function:*

$$\pi(Q) = R(Q) - C(Q) = (80Q - 2Q^2) - (20 + 10Q) = -2Q^2 + 70Q - 20$$

(b) *First-order condition (FOC):*

$$\frac{d\pi}{dQ} = -4Q + 70 = 0 \Rightarrow Q^* = 17.5$$

(c) *Maximum profit:*

$$\pi(17.5) = -2(17.5)^2 + 70(17.5) - 20 = -612.5 + 1225 - 20 = 592.5$$

(d) *Second-order condition:*

$$\frac{d^2\pi}{dQ^2} = -4 < 0 \Rightarrow \text{maximum holds.}$$

2. Utility Maximization Problem

Given:

$$U(x) = 4x - 0.5x^2$$

Solution:

(a) $\frac{dU}{dx} = 4 - x = 0 \Rightarrow x^* = 4.$

(b) Second derivative: $\frac{d^2U}{dx^2} = -1 < 0 \Rightarrow$ maximum.

(c) *Interpretation:* Utility increases with leisure up to 4 hours, then decreases due to diminishing satisfaction. Thus, $x^* = 4$ is the optimal leisure time.

3. Logarithmic Utility and Diminishing Marginal Utility

$$U(x) = \ln(x)$$

Solution:

(a) $U'(x) = \frac{1}{x}, \quad U''(x) = -\frac{1}{x^2}$

(b) $U'(x) > 0$ for $x > 0$ and $U''(x) < 0 \Rightarrow$ marginal utility is positive but diminishing.

(c) Elasticity:

$$E_{Ux} = \frac{U'(x) \cdot x}{U(x)} = \frac{1/x \cdot x}{\ln(x)} = \frac{1}{\ln(x)}$$

At $x = 10$, $E_{Ux} = \frac{1}{\ln(10)} \approx 0.434.$

Interpretation: Utility grows less than proportionally with x ; elasticity below one implies diminishing satisfaction.

4. Production Function and Marginal Product

$$Q = 10L^{0.5}$$

Solution:

(a) $MPL = \frac{dQ}{dL} = 5L^{-0.5} = \frac{5}{\sqrt{L}}$

(b) MPL declines as L increases, so diminishing marginal productivity begins immediately.

(c) Profit: $\pi = pQ - wL = 5(10L^{0.5}) - 2L = 50L^{0.5} - 2L$

FOC: $\frac{d\pi}{dL} = 25L^{-0.5} - 2 = 0 \Rightarrow L^{-0.5} = \frac{2}{25} \Rightarrow \sqrt{L} = 12.5 \Rightarrow L^* = 156.25.$

Maximum profit: $\pi(156.25) = 50(12.5) - 2(156.25) = 625 - 312.5 = 312.5.$

5. Maximization with Logarithmic Revenue

$$P(Q) = 100 - 4Q, \quad C(Q) = 20Q + 10, \quad R(Q) = P(Q) \cdot Q$$

Solution:

(a) $R(Q) = 100Q - 4Q^2$

$$\Pi(Q) = \ln(100Q - 4Q^2) - (20Q + 10)$$

(b) FOC:

$$\frac{d\Pi}{dQ} = \frac{100 - 8Q}{100Q - 4Q^2} - 20 = 0$$

Simplify numerator:

$$100 - 8Q = 20(100Q - 4Q^2)$$

$$100 - 8Q = 2000Q - 80Q^2$$

$$80Q^2 - 2008Q + 100 = 0$$

Solve quadratic:

$$Q^* = \frac{2008 \pm \sqrt{2008^2 - 4(80)(100)}}{160}$$

$$Q^* \approx \frac{2008 \pm 2006}{160} \Rightarrow Q^* \approx 25.08 \text{ or } 0.0125$$

The feasible value (ensuring $R > 0$) is $Q^* = 25.08$.

(c) *Interpretation:* Taking logs places higher weight on proportional revenue changes, approximating risk-averse or growth-sensitive behavior.

6. Elasticity and Logarithmic Differentiation

$$y = 20x^{0.8}$$

Solution:

Take logs:

$$\ln y = \ln 20 + 0.8 \ln x$$

Differentiate:

$$\frac{1}{y} \frac{dy}{dx} = 0.8 \frac{1}{x} \Rightarrow \frac{dy}{dx} = 0.8 \frac{y}{x}$$

Elasticity:

$$E_{yx} = \frac{dy}{dx} \cdot \frac{x}{y} = 0.8$$

Interpretation: A 1% increase in x raises y by 0.8%.

7. Application of Natural Logarithm Properties

$$\ln \left(\frac{x^3 y^2}{z^4} \right) = 3 \ln x + 2 \ln y - 4 \ln z$$

Interpretation: Output elasticity w.r.t. inputs can be read directly from coefficients. If x doubles (+100%), output increases by approximately $3 \ln 2 \approx 2.079$, holding y, z constant.